Alexander R. McAllister

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github.com/political-economy-analysis

EDUCATION

Brown University *June 2026–June 2027*
MPA, Social and Economic Policy (GPA 4.0)
James Madison University *August 2016–May 2020*
BS, Quantitative Finance and Mathematics. (GPA 3.27)

SUMMARY

Research Interests: inequality, housing, AI and labor dynamics, social and economic policy.
Programming: Stata, Python, R, Julia, AI coding assistants (Claude, Codex)
Skills: statistics for policy and program evaluation, time-series analysis, econometrics, differential equations and dynamic systems, financial economics.
Graduate Coursework: Statistics for Public Policy; Program Evaluation; Economics for Public Policy; Macroeconomics for Public Policy; Policy Analysis and Problem Solving; Applied AI for Public Policy; Dimensions of Social and Economic Policy; How Americans Decide
Undergraduate Coursework: Time Series Analysis; Intro to Probability and Statistics; Linear Algebra and Differential Equations; Analysis of Dynamic Differential Equations; Fourier Analysis and Partial Differential Equations; Computational and Numerical Algorithms; Calculus I/II/III, Advanced Calculus I; Intermediate Microeconomic Theory; Mathematical Finance; Financial Risk Management; Seminar in Financial Engineering

WORK EXPERIENCE

Freddie Mac *June 2019–April 2022*
Quantitative Analytics Associate II, Portfolio Strategy, Jan 2022–Apr 2022 (full-time)
Proposed a revision to the single-family capital framework as well as the Federal Housing Finance Agency's (FHFA) Enterprise Regulatory Capital Framework that adjusted the long-term house price forecast. The peak-to-peak/trough-to-trough method predicted strong house price growth and concretely implements the economic reasoning that housing markets can remain unresponsive to decreases in housing demand. Presented the methodology and policy recommendation to team leads and executives. Communicated on salient topics about the capital-risk requirements, the Government-Sponsored Enterprises' timeline for exiting conservatorship, its relation to FHFA regulation, and the implications of basing risk-assessments on persistent market failures in the housing market.
Single-Family Analyst, Loan Advisor and Strategy, May 2020–Dec 2021 (full-time)
Developed an hourly forecast for single-family loan submission volume that captures seasonality, productivity shifts through the week, and pent-up demand. Used to dynamically scale server capacity to handle volume increase uptime in the systems. Presented to executives both in SF and in technical departments during implementation. Directly advised executives on implementing and updating the model to fit ongoing changes in housing finance industry.
Single-Family Intern, Portfolio Strategy, June 2019–Aug 2019 (full-time)
Updated the post-disaster loan delinquency model with a wave-equation interpolation of monthly delinquency data, to measure disaster severity and the effected community's recovery ability.
Applied Consulting and Management (part-time) *August 2016–May 2020*
Operations Consultant for family business in aerospace government consulting.

RESEARCH

- National Science Foundation Research grant May–July 2017. "A Necessary Condition for Exponential Matrix Solutions of Differential Equations, 2018. With Dr. James Liu and Alfred Williams. College Mathematics Journal 45:1."
- Boosting Trading Algorithms with Data Filtration, 2020. Undergraduate capstone (Seminar in Financial Engineering)
- Academic presentations: 2018 Joint Mathematics Meetings; SUMS 2018; JMU Mathematics Colloquium.

TEACHING EXPERIENCE

- **IAPA 1201C Introduction to International Macroeconomics:** Public-Policy Graduate Fellow (Graduate Assistant) with Prof. Fabrizio Perri (Fall 2026)
- **M167 Topics in Mathematics:** Instructor, official one-credit introductory course in the JMU mathematics department (Fall 2018–Spring 2020)

LEADERSHIP AND ADDITIONAL ACTIVITIES

- Virginia State Collegiate Chess Champion, 2018; founder and president, JMU Chess (2018–2020)
- Vice President, Financial Algorithm Organization, a working group trading equities and derivatives (2018–2020)
- Producer, Math Connections Theater, an educational math and comedy theater production (2018–2019)

References available upon request.